

JASSER NAHALI

Software Engineer | Full Stack & Systems

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[Portfolio](#) | [GitHub](#)

PROFESSIONAL SUMMARY

Software Engineer and Computer Engineering student at ESPRIT University specializing in scalable full-stack architecture, microservices, and real-time backend engines. Proven track record of architecting low-latency synchronization protocols, containerized code execution sandboxes, and cloud-ready systems using TypeScript, Node.js, NestJS, Fastify, Spring Boot, and Docker.

TECHNICAL SKILLS

- **Software Engineering & Architecture:** RESTful API Design, Microservices Architecture, Real-Time Systems (WebSockets/Socket.io), Event-Driven Architecture, OOP, CI/CD
 - **Languages:** TypeScript, JavaScript, SQL, Java, C++, Python, HTML5, CSS3
 - **Backend & System Infrastructure:** Node.js, NestJS, Fastify, Express.js, Spring Boot, Docker, Docker Compose, Redis, VPS Deployment
 - **Databases & ORMs:** PostgreSQL, MySQL, Prisma ORM, Database Indexing & Query Optimization
 - **Frontend Systems:** React, Next.js, Angular, Ionic React, Tailwind CSS, TanStack Query, Framer Motion
 - **Testing & Quality Assurance:** Jest, React Testing Library, Postman, SonarQube, Git, GitHub Actions
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WORK EXPERIENCE

Software Engineer Intern | Medianet – Tunis, Tunisia

Jun 2026 – Aug 2026

Project: Code-Dual (Real-Time Competitive Coding Engine)

- **Architected low-latency duel synchronization protocol** using Socket.io and WebSockets (React, Fastify stack), maintaining sub-100ms state updates and live code state restoration across concurrent sessions.

- **Engineered secure multi-tenant code execution engine** utilizing containerized Docker sandboxes (Judge0) to process polyglot submissions (JavaScript, Python, C++, Java, Rust, Go) under strict CPU, memory, and timeout constraints.
- **Designed event-driven tournament engine** featuring single-elimination bracket generation, ELO-based seeding algorithms, automated forfeit routines, and real-time round progression.
- **Implemented enterprise security layer**, incorporating JWT with HTTP-only refresh tokens, Fastify rate-limiting, and PostgreSQL database models via Prisma ORM.

Software Engineer (Personal Project) | Nutriwell-Store

Present

Project: E-Commerce & Health Retail Platform

- **Designed full-stack e-commerce architecture** using NestJS and React, implementing transactional database operations, dynamic order ref generation, and real-time calculations.
- **Integrated geo-location engine** utilizing Leaflet and OpenStreetMap APIs to support real-time radius calculations, GPS mapping, and pharmacy availability filtering.
- **Configured multi-container Docker infrastructure** with automated SSH deployment pipelines via GitHub Actions to a remote VPS.
- **Optimized asynchronous state management** using TanStack Query to reduce redundant network overhead and ensure instantaneous UI updates.

Software Engineer | KiddySafe Monorepo Starter (Remote – Switzerland)

Jul 2025 – Aug 2025

- **Architected production-grade monorepo template** for location-aware family safety software utilizing Next.js, Ionic React, Node.js, Express, Prisma (PostgreSQL), and Docker.
- **Implemented real-time tracking pipelines** leveraging WebSockets, end-to-end encryption helpers, JWT session binding, and secure cookie storage.
- **Established comprehensive unit and integration testing frameworks** with Jest and React Testing Library to ensure high system reliability across micro-deployments.

Software Developer | Iprecision – Tunis, Tunisia

Jan 2022 – Jul 2022

Project: Field Intervention Management Platform

- **Engineered full-stack enterprise solution** handling 100+ operational interventions using Angular and Spring Boot REST microservices.
- **Decreased system response times by 20%** through Spring Boot backend optimizations and MySQL indexing strategies, ensuring 99.9% data availability.
- **Enforced software quality standards**, utilizing SonarQube static analysis to reduce code smells by 40% and resolving 95% of API issues via Postman test suites.

Software Developer | Siamo Startup – Tunis, Tunisia

Jan 2021 – Feb 2021

Project: Shop Management System

- **Engineered inventory tracking system** managing 200+ product entities with Spring Boot and MySQL.
 - **Reduced server processing overhead by 25%** by optimizing service layers and SQL data access patterns.
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EDUCATION

Engineering Degree in Web Technology & Cloud Computing

ESPRIT University – Tunis, Tunisia | 2024 – Present

License Degree in Computer Science Technology

ISSET Mahdia – Mahdia, Tunisia | 2020 – 2023

Bachelor’s Degree in Computer Science

Ibn Arafah High School – Tunisia | 2018 – 2019

LANGUAGES

- **Arabic:** Native
- **English:** Advanced / Proficient
- **French:** Intermediate